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(Key and Solutions for AIMCAT2421)

Key

SECTION – I

1. D
2. C
3. B
4. C
5. C
6. C
7. D
8. A
9. A
10. C
11. C
12. B
13. B
14. D
15. C
16. B
17. B
18. D
19. A
20. B
21. 1324
22. 3142
23. A
24. 2431

SECTION – II

1. 37
2. C
3. A
4. D
5. 72
6. A
7. C
8. 126
9. 18
10. B
11. B
12. A
13. A
14. D
15. C
16. B
17. 88083
18. B
19. 10478
20. A

SECTION – III

1. 45
2. 53
3. 155
4. A
5. D
6. D
7. C
8. 13
9. 0
10. 245400
11. B
12. B
13. C
14. A
15. B
16. C
17. D
18. D
19. 1252
20. 120
21. B
22. C

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 517

1. The author talks about combinatorics throughout the passage.
Option A: The fifth paragraph mentions that “talented mathematicians in all fields routinely use and invent combinatorial arguments”. Hence, this option is incorrect.
Option B: The fourth paragraph of the passage states that “Combinatorial arguments tend to ignore this knowledge” of mathematics. However, the passage does not state that they use knowledge which is “contradictory” to conventional knowledge. Hence, this statement is also incorrect.
Option C: The passage mentions that “Combinatorial arguments ***tend*** to ignore this knowledge”. It also states that combinatorics solves the problems in which “systematic developments of past knowledge are ***relatively useless*** in attaching present problems”. We cannot infer from this that combinatorics do not use ***any knowledge of mathematics.*** This option is extreme and is not the correct answer.
Option D: In the fourth paragraph, the passage mentions “Combinatorics can be attempted by inspired amateurs and dilettantes who know little of formal mathematics, and ***much of it was developed by such people***”. Hence, we can infer that combinatorics is developed mostly by people who are not professional mathematicians.
Therefore, the correct answer is option D.
Choice (D)

2. The author talks about the differences between combinatorial argument and knowledge-based argument in the third paragraph of the passage.
Option A: The example presented in the third paragraph talks about how the same problem can be solved using knowledge-based argument and combinatorial argument. Hence, this option is incorrect.
Option B: In the fourth paragraph of the passage, the author mentions that “Combinatorics therefore

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represents the areas of mathematics that are failures in this sense, those for which systematic developments of past knowledge are relatively useless in attacking present problems". Hence, there are problems which can be solved using combinatorial arguments but not by knowledge-based arguments. Therefore, this is not the correct answer.
Option C: From the fourth paragraph, we can infer that the scope of combinatorics includes those problems for which "systematic developments of past knowledge are relatively useless". Hence, combinatorial arguments can be used for solving such problems. Hence, this is the correct answer.
Option D: The author mentions that combinatorics were developed by amateurs and dilettantes. But by definition, understanding knowledge-based arguments requires prior knowledge of mathematics, which the layman will most probably not possess. Hence, this is not the correct answer.
Therefore, the correct answer is option C.
Choice (C)

3. The author mentions that traditional mathematicians spent "much of their lives learning lore about their subject" and "Combinatorial arguments tend to ignore this knowledge".
Option A: Combinatorics "represents the areas of mathematics that are failures in this sense, those for which systematic developments of past knowledge are relatively useless in attaching present problems.". However, we cannot infer from the passage that it ***highlights*** the failures of traditional mathematicians. Hence, this is not the correct answer.
Option B: Combinatorics tends to ignore the knowledge which mathematicians spent much of their lives learning. This is the primary reason why combinatorics was not popular among mathematicians. Therefore, this is the correct answer.
Option C: The author mentions that "One goal of formal mathematics is to systematize the structure of mathematical knowledge to abolish the need for intricate reasoning and thought". But we cannot infer from this that traditional mathematicians are not comfortable with intricate reasoning and thought.
Option D: While the passage mentions that much of combinatorics was developed by amateurs and dilettantes, we cannot infer from the passage that traditional mathematicians are not comfortable using this subject because of this reason. Hence, this is not the correct answer.
Therefore, the correct answer is option B.
Choice (B)

4. The author provides an example regarding the difference between knowledge-based argument and combinatorial argument. The same problem can be solved using knowledge-based argument or combinatorial argument.
Option A: It is given that the problem can be solved using the results of three theorems. A theorem by definition can be derived ad recreated. Hence, if a person, who is ignorant of the theorems, solves the problem by recreating these theorems, he would have, effectively, solved this problem using combinatorial argument. Hence, this is not the correct answer.
Option B: Since it is given that the problem can be solved using the results of the three theorems, it can be solved using knowledge-based argument (i.e., with the knowledge of the results of the three theorems). Hence, this option is also incorrect.
Option C: As mentioned in the explanation for option A, it can be solved using knowledge-based argument or by using combinatorial argument. Hence, this is the correct answer.
Option D: It is possible that true combinatorists also can solve the problem if they recreate the three theorems. Hence, this is not the correct answer.
Therefore, the correct answer is option C.
Choice (C)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 536

5. Option A: There have been changes in observational instrumentation and techniques over time. Such a conclusion doesn't need the assumption that there is necessarily ***an improvement*** in the quality of instrumentation and techniques over time. Hence, Option A is not the answer.
Option B: The sampling isn't reliable in the measurements of SSTs. No comments have been made or no conclusions have been made on the process of sampling and whether it is easy or difficult. Hence, Option B is not the answer.
Option C: Consider the second para of the passage. The author points out that the record of global surface temperature is produced by combining measurements of SST with measurements of air temperature over land and ice. The author then moves on to state that the uncertainties are associated with the SST estimates. So, it has been assumed that air temperatures over land have been measured reasonably accurately. Hence, Option C is the answer.
Option D: This is not needed for any of the arguments made in the second para. That is because the author simply states that patchy metadata complicates measurements. So, that statement cannot be arrived at if we didn't have any data about how data was collected in the past. Hence, Option D is not the answer.
Choice (C)

6. Option A: While the difference was introduced because of the shift in methodology, the author wasn't trying to highlight or give an opinion on whether Fahrenheit should be used or degree Celsius. That's because it isn't clear what's more inaccurate, if at all there is a difference. All it shows is that smallest of changes in measurement standards can produce huge deviations.

Option B: This is not way off the target looking at the text that precedes the example – '*Ships were grouped by national origin, on the assumption that ships from the same country would tend to have similar measurement practices at any given time. The authors found sizeable offsets in SST estimates between ship groups, ranging from −0.3 °C to +0.6 °C.*' This sets the context for the example and what it is trying to demonstrate. However, one must note that the example in the para uses the keyword 'also' to show that we are going to discuss another issue – that of approximation: '*Chan et al. [also] realized that measurements*'. So, differences between measurements of various countries wasn't the focus of this particular para, but the previous one. Hence, Option B is not the answer.
Option C: The para ends with '*Celsius and then dropping any numbers after the decimal point*'. That means the issue is because of dropping of the decimal point – approximation. It is because of that approximation that calculations went awry, the additional issue (covered in this para) apart from the fact that methodologies were different (and were introducing errors too). The 'also' in the starting line of the para is sufficient to identify this option. Hence, Option C is the answer.
Option D: The differences are between ship groups, probably from different nations and not from the same nation. This can be understood from '*Ships were grouped by national origin, **on the assumption that ships from the same country would tend to have similar measurement practices** at any given time.*' Hence, Option D is not the answer. Choice (C)

7. Option A: We cannot quite determine the geography of the oceans from the information given in the passage. That aspect of warming has not been discussed in the passage – just the inaccuracy of the measurements. Hence, Option A is not the answer.
Option B: Since it is mentioned that scientists have been '*...substantially underestimating warming in the North Pacific during the early twentieth*

the waters of the North Atlantic were warmer than they were originally thought to be (since the passage establishes that inaccurate measurements were made back then, and not not). Hence, Option D is the answer. Choice (D)

8. Option A: Chan et al. published a study that proved that the warming predictions are off because of the inaccuracies historically in measuring the temperature of the water in the early twentieth century. Ocean warming calculations are prone to inaccuracies in the methodologies and, the intrinsic climate variability itself has a smaller impact than was thought previously. Hence, Option A is the answer.
Option B: They are reasonably accurate as understood from '*These findings bring the difference in estimated warming between the two regions **in line with projections** from climate models...*' The findings of the study are in line with the climate model projections. So, the study has proved the veracity of the models. Option B is not the answer.
Option C: The study doesn't strengthen this argument because the study was about measurements from a century ago or further back in the past. The study doesn't doubt current measurements in any way. No information has been given in the passage about current methods of measurements or their cons. Hence, Option C is not the answer.
Option D: The study doesn't make a comment on global warming. It only suggests that in some cases, the warming of ocean waters has been overestimated, and in some cases, the warming has been underestimated. The warming itself is not a topic touched by the study. Hence, Option D is not the answer. Choice (A)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 529

9. The author's main argument has been highlighted

century…', we can say that the difference between current temperatures (fairly accurate) and past temperatures are bigger than they are thought to be. In other words, past temperatures recorded higher temperature than was the case. It means, the waters were less warm than they were thought to be. Hence, Option B is not the answer.
Option C: Climate variability doesn't have as big an impact, the para says. That doesn't mean that 'intrinsic' climate variability doesn't have any impact at all. It just has a smaller impact. Hence, Option C cannot be inferred from the last para.
Option D: Since the study suggests that the '***scientists have been overestimating warming in the North Atlantic***', we can understand that the warming isn't as much as what the scientists thought it was. In other words, the difference between the temperature then and now is not as high as scientists think it is. That's possible only if

in the last para. Consider the sentences: '*The integrated world's neglected problems have now grown in the eyes of the public to the point where the benefits of the global order are easily forgotten. Yet, the solution on offer is not really a fix at all. Slowbalisation will be meaner and less stable than its predecessor.*' The author intends to convey that globalisation has advantages and disadvantages. The former has been forgotten. Now, what has been seen as a replacement for the perils of globalisation – slowbalisation – is even worse. To refute it, we need an option that talks about slowbalisation not being worse than globalisation.
Option A: This statement, if true, shows that slowbalisation isn't really 'meaner' than its predecessor (globalisation). That goes against the author's main argument against slowbalisation – that it is worse than its predecessor, and also less stable. Option A is the answer.

Option B: Even if this were true, it wouldn't really affect the author's main argument in the passage in any way. This option talks about how slowbalisation came about. The author's main argument is about the consequences of the propagation of slowbalisation. Hence, Option B is not the answer.
Option C: By asserting that slowbalisation isn't the solution to the problem, this option tends to corroborate what the author was trying to derive in the passage – which is that slowbalisation is not a 'fix at all'. Hence, Option C is not the answer.
Option D: Even if this is true, the option doesn't highlight how slowbalisation will make things better, contrary to the disadvantages of slowbalisation that the author has listed out – like emerging countries bridging the gap with the right (their growth need not mean they will be able to bridge the gap), and the problems of climate change, migration and tax-dodging. Hence, Option D is not the answer. Choice (A)

10. Consider the following sentences: '*...slowbalisation will not fix the problems that globalisation created. Automation means that there will be no renaissance of blue-collar jobs in the West. Firms will hire unskilled workers in the* globalisation and were not given rise to by slowbalisation. Hence, Option D is not the answer.
Choice (C)

11. Consider the following statements: *Tax systems are being bent to patriotic ends…Trade is suffering as firms use up the inventories that they had built up in anticipation of higher tariffs...* ***But what really matters*** *is firms' long-term investment plans, as they begin to lower their exposure to countries and industries* ***that carry high geopolitical risk or face unstable rules****. There are signs that an adjustment is beginning. Chinese investment into Europe and America fell by 73% this year. The global value of cross-border investment by multinational companies [also] sank by about 20%.*
The author here starts with all the changes around the world and then arrives at what firms have been doing – which is to avoid investing in countries which are vulnerable to unpredictable changes, because of the local rules/governments.
Option A: The author isn't highlighting a ***decrease*** in global value of investment. In fact, firms are not investing in areas which are under erratic rules or regimes. Hence, Option A is not the answer.
Option B: While the author is discussing

cheapest places in each region. Climate change, migration and tax-dodging will be even harder to solve without global co-operation...' Here, the author indicates that contrary to popular opinion, slowbalisation will not fix problems ***created by globalisation***.

Option A: The author points out that the disadvantage of slowbalisation lies in how difficult it will become for emerging countries to bridge the gap with developed ones. ***A widening gap*** hasn't been mentioned anywhere in the passage. So, slowbalisation will make it more difficult for emerging countries to match/compete with the developed countries. That doesn't necessarily mean that emerging countries will regress. Option A is not the answer.

Option B: *Automation means that there will be no renaissance of blue-collar jobs in the West.* The author indicates that slowbalisation won't fix this issue. However, this is not a disadvantage of slowbalisation. The loss of blue-collar jobs was an adverse consequence of globalisation, which cannot be fixed by slowbalisation. Hence, Option B is not the answer.

Option C: This is the cause for turbulence in the slowbalisation era. That's because countries are affected by America's interest rates and yet, the trade links have been severed. So, there could be instability. This is a disadvantage of the slowbalisation era. Option C is the answer.

Option D: *Climate change, migration and tax-dodging* ***will be even harder*** *to solve without global co-operation.* That the author is saying that these problems will be even harder to solve, means that these problems existed prior to the advent of slowbalisation. These can be understood to be the disadvantages of

multinational companies, the idea is not to highlight the risks of all companies everywhere. It is not globalisation that is the risk but some countries, some rules and some regions. Hence, Option B is not the answer.

Option C: This is what the author is trying to say. Firms are looking at their long-term investments. They are ***being careful*** because in the long term they ***need to stay away from regions or countries which are unpredictable***. This fear of exposure is what is an indicator of slowbalisation. Option C is the answer.

Option D: Multinational companies not investing in some countries or regions is not entirely because they are scared by local companies. That could be one factor especially if the rules are in favour of local companies. But that is not the only factor covered, when the author says firms are being careful about their exposure. Changing tax regimes, for example, is another. Hence, Option D is not the answer.

Choice (C)

12. Consider the following: '*Slowbalisation will lead to deeper links within regional blocs. Supply chains in North America, Europe and Asia are sourcing more from closer to home. In Asia and Europe most trade is already intra-regional, and the share has risen since 2011. Asian firms made more foreign sales within Asia than in America this year. As global rules decay, a fluid patchwork of regional deals and spheres of influence is asserting control over trade and investment...*'

Option A: This has been mentioned directly in the line '*In Asia and Europe most trade is already intra-regional*.' Hence, the intra-regional trade has indeed increased. Option A is not the answer.

Option B: Consider the sentence: '*Most countries' interest rates will still be affected by America's even as their trade patterns become less linked to it, leading to financial turbulence...*' The author

improbability of true significance than on what the author considers limited achievement. Scepticism would be more appropriate. Choice D is not the answer.

Choice (B)

points out that even if trade connections and links are severed because of slowbalisation, there will be turbulence, because countries are still affected by the interest rates in America. That means, it can be inferred that even in the past, during globalisation, America's interest rates affected those of other countries. So, that will continue despite slowbalisation, according to the passage. Hence, Option B is the answer.

Option C: This can be understood directly from the sentence '*Supply chains in North America, Europe and Asia are sourcing more from closer to home*'. Option C is not the answer.

Option D: The author has established a correlation between the sluggish economy and the fact that costs of shifting goods in planes and ships have stopped falling. This can be understood from '*Globalisation has slowed from light speed to a snail's pace in the past decade for several reasons.* ***The cost of moving goods has stopped falling***.' Hence, Option D is not the answer. Choice (B)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 532

13. Option A: The author says that Asia, if at all it were to match the United States, would take very long to do so. However, the author is not optimistic about Asia's rise at all. Hence 'great fascination' is not the correct answer.

Option B: But ***it's a gross exaggeration*** to say that Asia will emerge as the world's predominant power player. At most, Asia's rise will lead to the arrival of a multi-polar world, not another unipolar one. The author mentions various constraints and obstacles as to why Asia cannot be a single entity of power, now or in the future. Hence, we can infer that the tone of the author towards the rise of Asia can be termed as one of strong scepticism. Choice B is correct.

Option C: While reiterating his belief that Asia cannot be a superpower, the author acknowledges the advancements made by Asia since World War II. (Sustained, rapid economic growth since World War II has undeniably boosted the region's economic output and military capabilities.) So 'pointed criticism' is not the tone of the author.

Option D: Sustained, rapid economic growth since World War II has undeniably boosted the region's economic output and military capabilities. Given Asia's relatively low per capita income, its growth rate will indeed outpace the West's for the foreseeable future. But the region faces enormous demographic hurdles in the decades ahead. The author mentions various other obstacles. Hence we cannot say that the tone of the author towards the rise of Asia can be termed as one of guarded appreciation. The focus is more on the

14. Statement (a): Pax Americana was made possible not only by the overwhelming economic and military might of the United States but also by a set of visionary ideas: free trade, Wilsonian liberalism, and multilateral institutions. Unlike America, Asia does not seem to play an equally inspiring role as a thought leader. So, statement (a) is a correct reason.

Statement (b): Statement (b) is not entirely true. Asia had a huge population and not America. Also 'dynamic economy capable of rising to the occasion when needed' may be true of any country, not only with the US. The second part of statement (b) has not been mentioned in the passage. (Although Asia today may have the world's most dynamic economies ....) So statement (b) is incorrect.

Statement (c): Asia is nowhere near closing its economic and military gap with the West. Pax Americana was made possible not only by the overwhelming economic and military might of the United States. ..... So, statement (c) is a correct reason.

Statement (d): Statement (d) has not been mentioned in the passage.

Only (a) and (c) are correct. Choice (D)

15. Option A: Those who think Asia's gains in hard power will inevitably lead to its geopolitical dominance might also want to look at another crucial ingredient of clout: ideas. Pax Americana was made possible by a set of visionary ideas: free trade, Wilsonian liberalism, and multilateral institutions. Although Asia today may have the world's most dynamic economies, it does not seem to play an equally inspiring role as a thought leader. .... The Asian export-dependent model of development will either disappear or cease to be a viable engine of growth. Hence choice A is true and is not the answer.

Option B: The fast ascent of one regional player will be greeted with alarm by its closest neighbors. China and Japan have fought repeatedly over Korea .... If Asia is becoming the world's center of geopolitical gravity, it's a murky middle indeed. Hence, choice B can be a factor impeding the rise of Asia as a superpower. Choice B is not the answer.

Option C: Given Asia's relatively low per capita income, its growth rate will indeed outpace the West's for the foreseeable future. Choice C by itself is not a hurdle or stumbling block in Asia's rise. So, choice C is the correct answer.

Option D: From "Environmental and natural resource constraints could also prove crippling. Pollution is worsening Asia's shortage of fresh water while air pollution exacts a terrible toll on health" and "the region faces enormous demographic hurdles in the decades ahead…", choice D is true and is not the answer. Choice (C)

16. Option A: Choice A is incorrect. Choice A is insufficient. The big idea animating Asians now is empowerment; Asians rightly feel proud that they are making a new industrial revolution. But self-confidence is not an ideology....

Option B: If Asia is becoming the world's center of geopolitical gravity, it's a murky middle indeed. Those who think Asia's gains in hard power will inevitably lead to its geopolitical dominance might also want to look at another crucial ingredient of clout: ideas. Pax Americana was made possible not only by the overwhelming economic and military might of the United States but also by a set of visionary ideas: free trade, Wilsonian liberalism, and multilateral institutions. Although Asia today may have the world's most dynamic economies, it does not seem to play an equally inspiring role as a thought leader, when it actually should, given the circumstances. Hence if Asia can address this lacuna, then it can rise to superpower status. Choice B is the answer.

Option C: Choice C is not enough. But self-confidence is not an ideology. .... The author does not actively advocate choice C in the passage. Hence choice C is not the answer.

Option D: The much-touted Asian model of development does not seem to be an exportable product. The Asian export-dependent model of development will either disappear or cease to be a viable engine of growth. But choice D has not been suggested in the passage. Choice (B)

Solutions for questions 17 and 18:

17. Option A: Art doesn't have a fixed meaning. This cannot be interpreted to mean 'art doesn't have a universal meaning.' Also, this sentence is totally different from saying art is a force against universalism. Also, this is not the reason why it should not be used for propaganda. Art cannot be used for activism, which is not the same as propaganda. So, Option A can be eliminated for several reasons.

Option B: This is apt as it represents both the main ideas of the para: that art doesn't have an end goal, which is why it cannot be used for activism, and that art extends the process of reflection ('a process of radical experimentation along a line of inquiry'). Inquiry here represents self-questioning/introspection/reflection. Option B is the answer.

Option C: The process of art evolves along a line of inquiry. This is different from saying art itself is evolving. Also, it is also incorrect to say that no artwork can have a comprehensive meaning. Such a thing has not been mentioned in the para. The option also misses out on the main idea of not using art for activism. Option C is not the answer.

Option D: While it is true that one cannot draw conclusions about art, that is not the reason why art cannot be used for activism. It cannot be used so because that is not the point of art. The point of art is different. It has its own meaning and that meaning manifests into artworks through experimentation along a line of inquiry. So, art cannot be used for activism because it cannot be something so specific as to have an end point (which is not the same as saying conclusions cannot be drawn from it). Hence, Option D is not the answer. Choice (B)

18. Option A: The para doesn't set up a comparison between Pacific islanders and others. Only the impact of climate change on Pacific islanders has been discussed. Option A is not the answer.

Option B: While this is partially true, the para's essence is not just about the Pacific countries' ocean dependency. They are dependent on land as well, and hence, are vulnerable to climate change. Option B is not the answer.

Option C: This is not the essence of the para because the para doesn't really carry a note of an ongoing devastation. These islanders are vulnerable, but saying that these countries are being devastated right now is not the intention of the para. Option C is not the answer.

Option D: This is apt as it mentions the two main ideas: Pacific islanders are especially susceptible to climate change (destitution and aid dependent are the only resort) and that is because of their dependence on land and ocean. Option D is the answer. Choice (D)

Solutions for questions 19 and 20:

19. The statement preceding blank (1) introduces the idea of Romanticism and how it evokes images of artists, poets, etc. The highlighted sentence talks about how different people perceive Romantics – some think they opposed reason and others think of candlelight dinners and declarations of love. The rest of the paragraph focuses on the Jena set and how the idea of Romanticism was more complex for them. The "though" in the statement succeeding blank (1) contrasts the notion in the first sentence of the paragraph as well as the highlighted sentence. Hence, the highlighted sentence fits in blank (1).

The statements preceding and succeeding blank (2) refer to the Jena set and how they viewed Romanticism. The highlighted sentence is generic and inserting it in blank (2) would break the flow. Hence, the highlighted sentence does not fit in blank (2).

The statements preceding and succeeding blank (3) are connected. The statement preceding blank (3) talks about how romantic poetry could merge different disciplines and subjects into something new. The statement succeeding blank (3) expands on this idea by imparting to the finite a shimmer of the infinite, i.e., combining two things. The highlighted sentence would break the flow. Hence, the highlighted sentence does not fit in blank (3).

The statement preceding blank (4) talks about how Novalis romanticizes and the statement succeeding it talks about Novalis's friends insisting that the romanticizing needs imagination. Therefore, the two sentences form a pair and hence, the highlighted sentence does not fit in blank (4). Choice (A)

20. The highlighted sentence must come after the statement following blank (1), which states that people enjoy the “extra” hour of daylight during the evening. The highlighted sentence counters this by saying that technically, setting the clocks one hour ahead does nothing to increase the amount of daylight. Hence, the highlighted sentence does not fit in blank (1).

The statement preceding blank (2) states that people enjoy the so-called extra hour of daylight during the evening and the highlighted sentence follows it up by saying that in reality, there is no extra hour. The statement following blank (2) states that the change in labels nudges us into waking up an hour earlier. This statement when preceded by the highlighted sentence sum up the idea of daylight saving by stating that even though the number of hours does not change, it can affect people’s behaviour. Hence, the highlighted sentence fits in blank (2).

The statements immediately preceding and following blank (3) talk about how daylight saving affects people’s behaviour. Inserting the highlighted sentence in blank (3) would break the flow as it talks about how daylight saving does not really increase the amount of daylight. Hence, the highlighted sentence does not fit in blank (3).

The statements preceding blank (4) talk about how changing the labels on the hours every day nudges people to do things and how it is perceived to be beneficial. The highlighted sentence devalues these benefits and cannot fit in blank (4) as it will disturb the flow of the para (all the positives that the para talks about will be weakened if the sentence is placed in blank 4). Therefore, it has to precede the statements preceding blank (4). Also, the statements following blank (4) talk about how Benjamin Franklin came up with the idea of daylight saving. The highlighted sentence is not directly related to these statements. Hence, the highlighted sentence does not fit in blank (4). Choice (B)

Solutions for questions 21 and 22:

21. Sentence 1 talks about the form in which mercury is present in wetlands – in sediments or in organic matter.

Sentence 2 talks about how mercury trapped in wetlands can be released. It starts with a contrast marker to, which means it is probably preceded by sentences that talk about mercury that is bound (for example, in Sentence 1).

Sentence 3 references another sentence with ‘this’ form – it also talks about a form from which ‘it’ (probably mercury) cannot escape. (It is to be noted that this sentence is also in contrast with Sentence 2.

Sentences 2 and 4 are connected because they talk about a different form of mercury, elemental mercury. Since, 2 starts with a contrast marker, 13 will precede 24. So, 1324 will be the para.
Ans: (1324)

22. Sentence 1 explains how captive birds can help in understanding social choices in gregarious (sociable) species.

Sentence 2 has a conclusion indicator ‘therefore’ (not necessarily conclusion of a para, but could be of an idea as well), but the big clue is ‘alternative’. So, we have to look for a method elsewhere in the para that doesn’t work.

Sentence 3 discusses why animals invest in social behaviour.

Sentence 4 talks about the difficulty of field-based measurements to gauge social networks.

42 is a block because 4 talks about a method that is not useful and 2 talks about a useful alternative. 1 is part of the same idea, but it must precede 4. That is because, if it doesn’t, then 12 will have to come together, which won’t make sense, because of the ‘therefore’ in 2 despite the fact that both sentences are saying similar things. So, 142 is the block. 3 is a good fit for the starting of the para. So, 3142 is the para. Ans: (3142)

Solution for question 23:

23. Option A: In multipolarity, trade-based tension is a possibility. So is a crisis of identity where countries are not part of a pole. So, is a crisis of ambition where countries want to, but cannot be a pole. This option mentions all the sub-ideas. Hence, Option A is the answer.

Option B: Multipolarity causes tensions of trade. However, the para doesn’t mention that there are also tensions between countries which do not want to align with the poles. We only know of countries that want to be a pole but cannot be or countries that don’t have a pole. So, countries wanting to be left out is not a sub-idea mentioned in the para. Option B is not the answer.

Option C: Disagreement in policy is not one of the factors involved in multipolarity. It is the very definition of multipolarity – the difference in policy language. So, this option misrepresents that piece of information. Option C is not the answer.

Option D: The first part of the option is not incorrect, in the sense that major forces disagree in policy leading to multipolarity. However, the para doesn’t mention that these forces leave out the other countries. That is not the essence of the para. Option D is not the answer. Choice (A)

Solution for question 24:

Sentence 4 again talks about a form of mercury where it can volatilize.
Sentence 1 has to be the introductory statement as it is the only independent statement. Sentence 1 and 3 are connected because in both cases, we are looking at a form of mercury that is 'trapped' (in 1) or 'can't escape' (in 4).

24. Sentence 1 needs a preceding sentence to establish what 'these systems' are. It is a positive sentence about these systems, essential as they are for any formulation of a quantum theory of gravity.
Sentence 2 is an independent, vague sentence that introduces closed timelike curves as controversial without explaining why.

Sentence 3 references a previous sentence that talks about paradoxes while also shifting the tone with 'However'.
Sentence 4 refers to 'they' – the legitimate solutions to Einstein's field equations, and explains the paradox.
We can understand that 24 is a block – one introduces the subject, the other introduces the paradox. 3 with a tone shift explains the resolution of the paradox. 1 is the last sentence. So, 2431 is the order. Ans: (2431)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	21
Medium	10, 11, 12, 14, 16, 18, 20, 22, 23
Difficult	1, 2, 3, 4, 5, 6, 8, 9, 13, 15, 17, 19, 24
Very Difficult	7

SECTION – II

Solutions for questions 1 to 5:

From the given conditions, we can see that, if all the five of them went for a walk, Abhinav walks for more distance than Jai, Kiran and Pavan. Also, Pavan walks for more distance than Tarun but Tarun walks for more distance than Abhinav. This is not possible. Hence, it is not possible for the five of them to go for a walk on any day.
Hence, exactly four persons must have gone for a walk during the given period.
Consider Abhinav did not go for a walk on a day.
If Tarun walked for x km, Pavan walked for x + 6 km, Kiran walked for x + 3 km and Jai walked for x + 5 km. Hence, the total distance that they walked will be 4x + 14 km.
Consider Jai did not go for a walk on a day.
If Abhinav walked for x km, Tarun walked for x + 5 km, Pavan walked for x + 11 km, Kiran walked for x + 8 km. Hence, the total distance that they walked will be 4x + 24 km.
Consider Kiran did not go for a walk.
If Jai walked for x km, Abhinav walked for x + 1 km, Tarun walked for x + 6 km, Pavan walked for x + 12 km. The total distance that they walked will be 4x + 19 km.
Consider Pavan did not go for a walk.
If Kiran walked for x km, Jai walked for x + 2 km, Abhinav walked for x + 3 km, Tarun walked for x + 8 km. The total distance that they walked will be 4x + 13 km.
Consider Tarun did not go for a walk.
If Kiran walked for x km, Jai walked for x + 2 km, Abhinav walked for x + 3 km, Pavan walked for x + 3 km. The total distance that they walked will be 4x + 8 km.
For Day 1, if Abhinav did not go for a walk, then $4x + 14 = 27 \Rightarrow x = 3.25$ km. This is not possible because everyone walked for an integral number of km. Similarly, we can subtract 24, 19, 13 and 8 from 27 and check which of those will be divisible by 4.
On Day 1, the only possibility is that Kiran did not go for a walk. In this case, x = 2.
Similarly, on Day 2, Abhinav did not go for a walk and x = 3.
On Day 3, Pavan did not go for a walk and x = 4.
On Day 4, Abhinav did not go for a walk and x = 6.
On Day 5, Pavan did not go for a walk and x = 2.
On Day 6, Tarun did not go for a walk and x = 3 (For Jai, the value of x will be a negative integer which is not possible).
Similarly, we can find the values for the other days as well.
The following table provides the distances walked by each person on each day (with 'X' representing that person not going for a walk on that day):

Day	Abhinav	Jai	Kiran	Pavan	Tarun
Day 1	3	2	X	14	8
Day 2	X	8	6	9	3
Day 3	7	6	4	X	12
Day 4	X	11	9	12	6
Day 5	5	4	2	X	10
Day 6	6	5	3	6	X
Day 7	5	4	2	5	X
Day 8	4	3	X	15	9
Day 9	7	6	4	X	12
Day 10	X	10	8	11	5

1. The total distance that Abhinav walked across the ten days is 37 km. Ans: (37)

2. Jai went for a walk on all the ten days during the given period. Choice (C)

3. The required difference = 65 – 59 = 6 km. Choice (A)

4. Pavan went for a walk for a maximum of three consecutive days (Day 1 – Day 2, Day 6 – Day 7, Day 7 – Day 8). Choice (D)

5. Pavan walked for the maximum distance of 72 km. Ans: (72)

Solutions for questions 6 to 10:

Given that among the teams that played exactly two matches, one team scored the same number of runs in both the matches that it played. Among the runs scored by each team, only for E is the number of runs scored even. Hence, E must have played exactly two matches and scored 142 runs in each match that it played.
Similarly, among the runs scored against each team, only for D is the number of runs scored against even. Hence, D must have played exactly two matches and against D, both the teams must have scored 142 runs each.
Since D and E played two matches each and A and B could not have played one match (from (iii)), C must have played only one match.
C could not have played against D or E because C must have scored 271 runs and the opposite team must have scored 263 runs in that match.
Hence, C must have played against A or against B.
In the match with D, A scored 142 runs and D scored 147 runs. Hence, A would have lost this match.
In the match with E, A scored 119 runs and E scored 142 runs. Hence, A would have lost this match.
Therefore, this case is not possible, from (iv), since A lost two matches and won only one match.

Case 2: Consider that C played against A.
In the match between C and A, C must have scored 271 runs and A must have scored 263 runs.
The number of runs scored by A in the other matches must be 389 – 263 = 126. From (iii), A could have played only one more match, while B must have played three matches.
The number of runs scored against A in the other match must be 407 – 271=136.
A could not have played any match against D or E since A could not have scored 142 runs and the opposite team also could not have scored 142 runs.
Hence, A must have played against B. In this match, A scored 126 runs and B scored 136 runs. This match must have been won by B.
Since A played against B and B played three matches, B must have played against both D and E.
The last match can only be between D and E.
Since A scored 126 runs against B, the other teams must have scored 399 – 126 = 273 runs against B. Of these, E would have scored 142 runs against B. Hence, D must have scored 273 – 142 = 131 runs against B. Since D scored a total of 267 runs in the two matches that it played, D must have scored 267 – 131 = 136 runs against E. Since D scored 136 runs against E, B must have scored 239 – 136 = 103 runs against E. Against D, B must have scored 142 runs.
B would have won the match against D (since B scored 142 runs and D scored 131 runs). B would have lost the match against E (since B scored 103 runs and E

Case 1: Consider that C played against B.

In the match between B and C, C would have scored 271 runs and B would have scored 263 runs.

Since B scored a total of 381 runs, B must have scored 118 runs in the remaining match. From (iii), B could have played only one more match. Since B played only two matches, A must have played three matches.

The total runs scored against B is 399. Since C scored 271, the other team must have scored 128 runs.

In the match, where B scored 118 and the opposite team scored 128, B could have played against neither D nor E. Hence, this match must be against A. And A would have won this match as it scored more than B.

The runs scored against A is 407. Of these, B scored 118. Hence, the other teams must have scored 289.

The runs scored by A is 389. Of these, A scored 128 against B. Hence, A must have scored 261 in the other two matches.

Since A played two more matches, A must have played against both D and E. Also, D and E must have played against each other.

In the match against D, A must have scored 142 runs. Hence, in the match against E, A must have scored 261 – 142 = 119 runs.

In the match against E, E must have scored 142 runs. Hence, in the match against D, D must have scored 407 – 142 – 118 = 147 runs.

scored 142 runs).

The following table provides the details of the matches and the runs scored:

Runs Scored	Team 1	Team 2	Runs Scored
126	A	B	136
263	A	C	271
142	B	D	131
103	B	E	142
136	D	E	142

6. The runs scored by D in the two matches are 131 and 136. Hence, option A is the correct answer. Choice (A)

7. Both C and E won all the matches that they played. Choice (C)

8. The lowest score made by A, D or E in a match was 126 (by A). Ans: (126)

9. Given that the runs scored by and against each team became the same. In the first two matches that A played, A scored 18 runs less than that of the opposition teams. Hence, in the next matches, A should score 18 runs more than the opposition teams. Ans: (18)

10. Given that at least one match played by C was tied. Also, the runs scored by and against each team became the same. Since B has to play only one match (against C), in that match B must have scored 18 runs more than C. Hence, C could only have tied against D or E.

Consider that the match between C and D is a tie. In that case, in the match against E and C, C must have scored 10 runs more than E (for the runs for and against C to be the same).

In the match between A and E, A must have scored 35 runs against E (for the runs for and against E to be the same). In the match between A and D, D must have scored 17 runs more than A (for the runs for and against D to be the same).

The following table provides the runs scored by the two teams in the remaining matches of the tournament (taking 0 as the base for one team):

Runs Scored	Team 1	Team 2	Runs Scored
+18	B	C	0
0	D	C	0
0	E	C	+10
0	A	D	+17
+35	A	E	0

Now, consider that the match between C and E be a tie. In that case, in the match against D and C, C must have scored 10 runs more than D (for the runs for and against C to be the same).

In the match against A and E, A must have scored 45 runs against E (for the runs for and against E to be the same). In the match between A and D, D must have scored 27 runs more than A (for the runs for and against D to be the same). This is the second possible case, as mentioned below.

Runs Scored	Team 1	Team 2	Runs Scored
+18	B	C	0
0	D	C	+10
0	E	C	0
0	A	D	+27
+45	A	E	0

From the two cases, we can see that, in the match between A and D, the difference in runs is either 17 or 27. Choice (B)

Solutions for questions 11 to 15:

Let 1 to 7 represent the seats from left to right in which they were seated before the interviews began. The person at 7 is interviewed 1st, the person at 1 is interviewed 2nd, the person at 6 is interviewed 3rd and so on.
From (i), John was interviewed before Bob. Also, since John was at an extreme end only after Andy's interview, John must have been interviewed after Andy. John will not be in position 1 or 7.
From (ii), Tom and Will must be seated in positions such that the number of persons between them is the same as the number of persons interviewed after one of them and before the other. By checking different combinations of seats, the only possibilities for this are if they are seated in positions 3 and 4 and were interviewed 6th and 7th OR if they are seated in positions 3 and 6 and were interviewed 6th and 3rd.

Case 1: Tom and Will are in positions 3 and 4
Tom must be in position 3 and Will must be in position 4 (since the condition mentions about the number of persons interviewed after Tom and before Will).
Tom and Will would be interviewed 6th and 7th respectively. From (iii), Ben must be in position 5 (since he has to be adjacent to Will). John was interviewed before Bob. Since John is not the 1st or 2nd person to be interviewed, John will be in position 6 (interviewed 3rd) and Bob will be in position 2 (interviewed 4th). In this case, Harry cannot be at position 7, since he was not the first person to be interviewed. Harry cannot be at position 1 as well, since he was not sitting adjacent to Bob. Since Harry cannot be in any of the remaining positions, this case is not possible.

Case 2: Tom and Will are in positions 3 and 6
Tom must be in 6 (interviewed 3rd) and Will must be in 3 (interviewed 6th). Ben must be in position 5. Since John is not at position 1 or position 7, John must be in position 2 (interviewed 4th) and Bob must be in position 4 (interviewed 7th). Since Harry is not the first person to be interviewed, Harry must be in position 1 (interviewed 2nd) and Andy must be in position 7 (interviewed 1st). This is the only possible case.

The following table provides the order in which they were seated and the order in which they were interviewed:

Seating Order	1	2	3	4	5	6	7
Person	Harry	John	Will	Bob	Ben	Tom	Andy
Interview Order	2	4	6	7	5	3	1

11. Two persons (John and Will) are sitting between Harry and Bob. Choice (B)

12. John was interviewed immediately before Ben. Harry was sitting to the immediate left of John. Choice (A)

---

13. Will was sitting to the right of John. Ben was interviewed immediately before Will. Choice (A)

14. Will and Bob were sitting adjacent to each other and were interviewed one after the other. Choice (D)

15. Bob was the last person to be interviewed. Choice (C)

Solutions for questions 16 to 20:

Hence, the minimum possible value of N is 10478. Ans: (10478)

20. Let 0 be part of N. The next digits can be 3, 6, 7. There cannot be any more digits.
Let 0 not be a part of N. The digits can start from 2. The next digits can be 3, 6, 7. Even then, it is not possible to form a five-digit number.
Hence, it is not possible to form a five-digit number. Choice (A)

16. Given that the second digit of N is five more than the first digit. Hence, the first digit can be a minimum of 1 (as it cannot be 0). The second digit can be a minimum of 6.
The fourth digit must be 9. This is the only possible combination of these three digits.
Since 1 is the first digit, 3 must not be in N.
Since 6 is present in N, 4 and 8 cannot be in N.
Since 9 is present in N, 7 cannot be in N.
Hence, only 0, 2 and 5 can be in N.
If 0 is the third digit, the fifth digit must be 5 and N = 16095.
If 2 is the third digit, the fifth digit must be 5 and N = 16295.
If 5 is the third digit, the fifth digit can be 0 or 2. N can be 16590 or 16592.
Hence, a total of 4 possibilities exist for N.
Choice (B)

17. To find the minimum possible value of N, we can assume that 1 and 0 are in N. 2 and 3 cannot be in N. Let 4 and 5 be in N. The last digit can be 8.
For these digits, the minimum value of N = 10458.
This is the minimum possible value of N.
To find the maximum possible value of N, we can assume that 9 and 8 are in N. 7 and 6 cannot be in N. Let 5 and 4 be in N. The last digit can be 1.
For these digits, the maximum value of N = 98541
Required difference = 98541 – 10458 = 88083
Ans: (88083)

18. Given that 0 is not in N. Let 1 be one of the digits. If the second digit is 2, then the third and fourth can be 5 and 6. The last digit must be 9.
Hence, 4 need not always be a part of N.
Let the first digit be 1. If the second digit is 4, then the third and fourth digits can be 5 and 7. In this case, the last digit must be 8. Hence, 6 need not always be a part of N.
Consider 5 is not in N. If two digits are 2 and 3 (or 1 and 2), then the other two digits can be 6 and 7. However, we cannot form a five-digit number in this case.
If 5 is not a part of N, then we cannot form a five-digit number.
Therefore, 5 must be a part of N.
Choice (B)

19. 5 is not in N. Let 0 and 1 be two digits. 2 and 3 cannot be part of N. 4 can be the next digit. 5 and 6 cannot be part of N. 7 and 8 can be the next digits.

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	16, 17, 19
Medium	11, 12, 13, 14, 15, 18, 20
Difficult	1, 2, 3, 4, 5, 6, 7, 8
Very Difficult	9, 10

SECTION – III

Solutions for questions 1 to 7:

1. Considering $\frac{1}{3} - x$, we get,
Let $S = 1 - 4x + 9x^2 - 16x^3 + 25x^4 - 36x^5 + \ldots$ where $x = \frac{1}{3}$.

$$\therefore Sx = \quad x - 4x^2 + 9x^3 - 16x^4 + 25x^5$$

$$S(1+x) = 1 - 3x + 5x^2 - 7x^3 + 9x^4 - 11x^5 + \ldots$$

$$xS(1+x) = x - 3x^2 + 5x^3 - 7x^4 + 9x^4 - \ldots$$

$$S(1+x)^2 = 1 - 2x + 2x^2 - 2x^3 + 2x^4 - 2x^5 + \ldots$$
$$S(1+x)^2 = 1 - 2x\{1 - x + x^2 - x^3 + \ldots\}$$
$$S(1+x)^2 = 1 - \frac{2x}{1+x}$$
$$S(1+x)^2 = \frac{1-x}{1+x}.$$
$$\therefore S = \frac{1-x}{(1+x)^3}.$$

Now, substituting $x - \frac{1}{3}$, we get

$$S = \frac{1-\frac{1}{3}}{\left(1+\frac{1}{3}\right)^3} = \frac{2}{3} \times \frac{3^3}{4^3} = \frac{9}{32}.$$

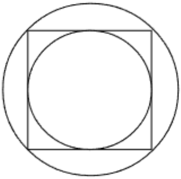
Therefore, $160S = 160 \times \frac{9}{32} = 45$

Ans: (45)

2. The least value of x and y, that satisfy the equation $5x - 19y = 2$ are $x = 8$ and $y = 2$
The next values of x that satisfy the equation are $8 + 19, 8 + 38, 8 + 57 \ldots\ldots, 8 + (19 \times 52)$
Total number of values = 53 Ans: (53)

3. Let the marks of the students in decreasing order be a, b, c, d and e.
$a + b + c + d + e = 134 \times 5 = 670$
$a + b + c = (134 + 23) \times 3 = 471$
$c + d + e = (134 - 16) \times 3 = 354$
$c = 471 + 354 - 670 = 155$. Ans: (155)

4.



Let the radius of the outer circle and the inner circle be R and r respectively
Side of the square = Diameter of the inner circle = 2r
Diagonal of the square = Diameter of the outer circle = 2R

From this, side of the square = $\frac{2R}{\sqrt{2}}$

$\therefore 2r = \frac{2R}{\sqrt{2}}$

$\Rightarrow \frac{R}{r} = \frac{\sqrt{2}}{1}$ Choice (A)

5.

A	B	C
200	196	?
	200	190

When B runs 196m, C runs $\frac{196 \times 190}{200} = 186.2$ m

Hence, A beats C by 200 – 186.2 = 13.8m
Therefore, if A and C finish the race at the same time, C should get a headstart of 13.8m
Choice (D)

6. The product of all the factors of
$N = N^{(\text{No. of factors of N})/2} = N^8$
Therefore, the number of factors of N is 16
Now, checking for the number of factors of the number given in each option, we get (A) → 12; (B) → 12; (C) → 14 and (D) → 18. Since (D) has 18 factors, it cannot be a factor of N.
$\therefore$ Among the options, only (D) cannot be a factor of N. Choice (D)

7. Let the cost price of the article be denoted by x.

Now $x + \frac{25}{100}x = 120$

$\Rightarrow \frac{5}{4}x = 120$

$\Rightarrow x = 96$.
In order to make a profit of 50%, he should have sold it for $x + \frac{50}{100}x = \frac{3}{2}(x) = \frac{3}{2}(96) = 144$.
Choice (C)

Solutions for questions 8 and 9:

There are 128 students in the class with 100 + 96 + 99 + 90 i.e., 385 passes. It is also known that 40 students passed in 4 subjects. So, they have a total of 160 passes. The remaining 88 students got the remaining 225 passes. Now, each of these 88 students can get at most 3 passes.

8. If the number of students who have no passes has to be maximised, then the number of students who have at least one pass has to be minimised. The number of students who have at least one pass will be minimum, when each student gets maximum number of passes i.e., 3 passes (because the students getting 4 passes are already taken care of). $\therefore$ The minimum number of students required to get the 225 passes = $\frac{225}{3} = 75$

i.e., each of these 75 students have exactly three passes and there are no students with one pass or two passes.
$\therefore$ Number of students with no passes
= 128 – (40 + 75) = 13 Ans: (13)

9. In the above problem, we could see that the number of students with two passes is zero. (the minimum possible) Ans: (0)

Solutions for questions 10 to 22:

10. Let us denote the n^{th} term in the series by T_n.
Now, $T_{59} = 15$

$T_{60} = \frac{20}{T_{59}} = \frac{20}{15} = \frac{4}{3}$

$T_{61} = \frac{20}{T_{60}} = \frac{20}{\frac{4}{3}} = 15$

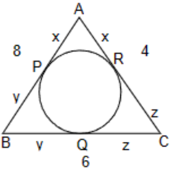
$\therefore$ Each odd term is 15 and each even term is $\frac{4}{3}$.

Thus, the required sum, from 278th term to 378th term =
S = sum of 51 even terms + sum of 50 odd terms

$= 51 \times \frac{4}{3} + 50 \times 15 = 68 + 750 = 818$.

$\therefore$ 300S = 818 × 300 = 245400. Ans: (245400)

11.



AB + BC + CA = 2(x + y + z)

$x + y + z = \frac{8+6+4}{2} = 9$

$\therefore$ x = (x + y + z) – (y + z) = 9 – 6 = 3
z = (x + y + z) – (x + y) = 9 – 8 = 1
x : z = 3 : 1 Choice (B)

12. $2x - 7y + t = 0$ and $5x - 3y - 2t = 0$
$\Rightarrow t = 7y - 2x$ $\Rightarrow t = \frac{5x - 3y}{2}$

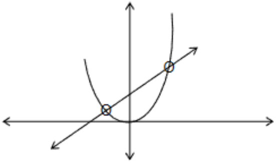
Equating *t*, we get

$7y - 2x = \frac{5x - 3y}{2}$

$14y - 4x = 5x - 3y$
$17y = 9x$

$\therefore \frac{x}{y} = \frac{17}{9}$ Choice (B)

13. The equations of the two curves are $y_1 = x^4 + x^2 + 5$ and $y_2 = x^2 + x + 6$

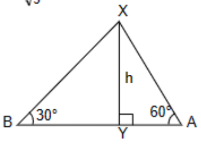


At the point of intersection $y_1 = y_2$
$\Rightarrow x^4 + x^2 + 5 = x^2 + x + 6 \Rightarrow x^4 = x + 1$
This is a quartic (4th degree equation) equation and finding the roots for this can be really complex. Therefore, we'll use graphs to find out the number of points of intersection.
The graphs of x^4 and x + 1 are given above. We see that there are two points of intersection.
Choice (C)

14. When an arithmetic progression, none of the terms of which are zero, has sum of its terms as zero, there are equal number of terms above zero and below zero. Hence, it always has an even number of terms. Choice (A)

15. Let the height of the tower (YX) be *h*.

$\therefore AY = \frac{h}{\sqrt{3}}$ and $BY = h\sqrt{3}$.



Remaining distance to be covered by
$B = h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$.

B will cover the remaining distance in another 15 × 2
= 30 minutes. Choice (B)

16. The given expression is

$$P^2\left(\frac{1}{R^2}+\frac{1}{Q^2}\right)+Q^2\left(\frac{1}{P^2}+\frac{1}{R^2}\right)+R^2\left(\frac{1}{Q^2}+\frac{1}{P^2}\right)$$

$$-\frac{P^2}{R^2}+\frac{R^2}{P^2}+\frac{P^2}{Q^2}+\frac{Q^2}{P^2}+\frac{R^2}{Q^2}+\frac{Q^2}{R^2}$$

As A.M > G.M for distinct positive numbers, each of

$\frac{P^2}{R^2}+\frac{R^2}{P^2}, \frac{P^2}{Q^2}+\frac{Q^2}{P^2}$ and $\frac{R^2}{Q^2}+\frac{Q^2}{R^2}$ is more than 2.

$\therefore$ The given expression must be more than 6.
Choice (C)

17. Since they together took 45 minutes to do the work that Bulu alone would have done in one hour, together they are $\frac{60}{45} = \frac{4}{3}$ times as efficient as Bulu.

$a = \frac{b}{3}$ (where, *a* and *b* are their rates of doing work)

Had they worked for equal intervals of time they would have shared the amount in the ratio of their rates of work.
$\therefore$ Bulu's share would have been

$\frac{3}{(1+3)}$ (6000) = ₹4500 Choice (D)

18. Choice (A): Each of x^2y, y^2x and z^2x is odd. Hence their sum is odd, which, in turn, when multiplied by *xyz* will result in an odd number.
$\therefore$ (A) is true
Choice (B): $(xy + yz + zx)$ is odd and $(x + 2y + 3z)$ is even. Therefore, their product is even.
$\therefore$ (B) is true.
Choice (C) : $(xy + yz + zx)$ and $(2x + y)^2$ are odd. Hence their product is also odd.
$\therefore$ (C) is true.
Choice (D): $(2x + 3y + 4z)$ is odd and $(xyz + 6)$ is also odd. Hence their product is odd.
$\therefore$ (D) is not true. Choice (D)

19. $(40)_8 = (32)_{10}$
$\Rightarrow (x)_{10} = (320)_{10}$
$\Rightarrow (x)_6 = (1252)_6$ Ans: (1252)

20. If n is the number of men, then number of matches between men and the number of matches between men and women will be ${}^nC_2$ and 4n respectively.

Therefore, $\frac{n(n-1)}{2} - 4n = 18$

Since A and B are travelling at the same speed, in the time A covered $\frac{h}{\sqrt{3}}$ to reach Y, B will also cover $\frac{h}{\sqrt{3}}$

$\Rightarrow n^2 - 9n - 36 = 0$
$\Rightarrow (n - 12)(n + 3) = 0 \quad \Rightarrow n = 12$
Total number of people in the group = 12 + 4 = 16.
Hence number of games played = 16(15/2) = 120.

Ans: (120)

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21. The equations of the lines are as follows.
$5x + 12y + 21 = 0$
$5x + 12y + 99 = 0$

The distance between them is $\left|\frac{99-21}{\sqrt{5^2+12^2}}\right| = \frac{78}{13} = 6$

Choice (B)

22. Any number with two distinct digits will be of the form *aabb* or *abab* or *abba* or *aaab* or *abaa* or *aaba* or *baaa*

Case I
If *aabb* or *abab* or *abba* is to be divisible by 9, then $2a + 2b$ must be either 9 or 18 or 27 or 36.
$2a + 2b$ cannot be 9 or 27 because it must be even.
Hence, if $2a + 2b = 18$, then (a, b) can be (1, 8), (2, 7),
(3, 6), (4, 5), (9, 0).
For each possibility of (a, b) except (9, 0), there will be 6 numbers possible.
For (9, 0) there will be three possible numbers.
Hence, there are a total of 4 × 6 + 3= 27 possibilities.
If $2a + 2b = 36$, then $a + b = 18$. This is only possible if
$a = b = 9$. Since this does not have two distinct digits, there are no possibilities.

Case II
If *aaab* or *abaa* or *aaba* or *baaa* is to be divisible by 9, then $3a + b$ can be 9 or 18 or 27 or 36.
If $3a + b = 9$, then (a, b) can be (0, 9), (1, 6), (2, 3), (3, 0).
If $3a + b = 18$, then (a, b) can be (3, 9), (4, 6), (5, 3), (6, 0).
If $3a + b = 27$, then (a, b) can be (6, 9), (7, 6), (8, 3), (9, 0).
If $3a + b = 36$, then (a, b) can be (9, 9) (not possible).
For each value of (a, b) except for (0, 9), (6, 0), (9, 0),
(3, 0), there will be four possible numbers.
For (0, 9) only one value is possible (9000).
For (6, 0), three values are possible.
For (9, 0), three values are possible.
For (3, 0), three values are possible.
Hence, the total number of possibilities are 8 × 4 + 10 = 42
Total possible numbers = 42 + 27 = 69.

Choice (C)

Difficulty level wise summary - Section III

Level of Difficulty	Questions
Very Easy	7
Easy	4, 12, 14, 18, 19, 20, 21
Medium	2, 3, 5, 10, 11, 15, 16, 17
Difficult	6, 8, 9, 13
Very Difficult	1, 22

Done

Tip: Scroll slowly for the best results